Name.....

Signature……..**Candidates Personal Number**…….

545/1

CHEMISTRY.

Paper 1.

Aug /2025.

2 hours

CHEMISTRY DEPARTMENT

New Lower Secondary Chemistry Examination

Uganda Certificate of Education

Internal mock set 7

Paper 1

2 hours

INSTRUCTIONS:

This paper consists of **two** *sections* **A** *and* **B**. *It has* **six** *examination items.*

Section **A** *has* **one compulsory** items.

Section **B** *has* **two parts, I** and **II**. *Answer* **one** item from each part.. Answer **four** items in all.

Answers to section **A must** *be written in the spaces provided while those of section* **B must** *be written on the provided answer sheets.*

Any additional item(s) answered will **not** *be scored*

SECTION A

Answer **all** *items in this section in the spaces provided.*

Item 1

Mariam, a small scale investor dealing in local juice production is facing a significant challenge. Her fruit juice, while praised for the delicious flavourful taste, the customers are complaining that it has a thin mouthfeel leading them to believe it's diluted. Further more , they are frustrated by the rapid settling of the fruit pulp ,which forces them to constantly shake the bottle to get a consistent product. She has come to you for guidance to address these issues to maintain her customer base in the competitive market.

Task. As a chemistry student,

(a) Categorize the products she should use to ensure quality production.

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(b) State how the products will ensure customers satisfaction.

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(c) advise her on that dangers associated with use of the products

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(d) evaluate the products used in ensuring customers satisfaction.

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Item 2.

Anthony, is an investor who intends to set up a manufacturing company, whose production process will involve converting many **ethene** molecules by *addition polymerization* to a **useful** material as equation below.

$$nCH_2{=}CH_2 \rightarrow \left[CH_2CH_2\right]_n$$

However, he is interested to know more about the reaction and the qualities of the material for various applications which can allow him to adapt to market demand while ensuring that his clients are well informed of environmental responsibility. He has consulted you for guidance.

Task: As a Chemistry student,

(a) explain to him the category of the **hydrocarbon** from which the material is produced.

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(b) guide him to understand the suitability of the material to its application in life.

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(c) advise him about the environmental impact associated with use of the material so as he can guide his clients appropriately.

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SECTION B

Part I (*Attempt one item from this section)*

Item 3 Due to high demand of the large quantities of nitric acid on market needed for production of ammonium nitrate fertilizer, the government has cleared an investor to set up a nitric acid manufacturing plant city. However ,the locals are concerned about its impact and are seeking assurance on how they will benefit from it's operations. The city officials have organized a workshop to educate the locals. You have been invited to attend.

Task.

As a chemistry student, make a write up you will use during the presentation.

Item 4.

Uganda is actively pursuing the development of its first oil refinery to ensure domestic energy security. The refinery is planned to be built in kabaale village, Buseruka Sub county in Hoima district near the rich crude oil fields. However, residents are concerned about the potential for environmental damage from refinery process and concerned about how they will directly benefit from the plant's operations. The official from ministry of energy have organized workshops to address the locals . You have been invited to address the locals .

Task. Write a write up presentation you will use during the presentation.

Part II

. **(*Attempt one item from this part.*)**

Item 5. The **Nile river** and the **surrounding environment** are facing a severe crisis. The once clean waters and surrounding environment are now degraded by a rapid population growth and industrialization which have led to devastating environmental consequences .These issues have created an urgent crisis, raising alarm among environmentalists . They have raised effort to develop and implement a joint action plan aimed at educating the public and mitigating these effects by organizing radio talk shows to create awareness to the public. You have been invited to address the public.

Task.

As a chemistry prepare a short presentation you will deliver in the radio talk show

Item 6.

Uganda discovered **reserves of natural gas** found in the fields of Albertine region.. Large volumes of gas will be transported primarily through pipelines using network of high-pressure steel pipes connecting production sites to processing plants, storage facilities and eventually to customers. However, this has raised significant alarm among public where pipelines will pass and are worried about the potential consequences . The government has organized workshops to raise awareness amongst the locals living in such areas. You have been invited to attend one of the workshops and address the locals.

Task. As a chemistry student, prepare a short message you will use to deliver in the workshop.

End.